

EMMM, Explain Me My Model! Explainable Multi-Dimension Multi-Level Multi-Strategy Chatbot Detection

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Abstract

The rapid advancement of large language models (LLMs) introduces new security risks to online emergency and customer service platforms, where malicious actors can exploit LLMs to conduct large-scale user impersonation through AI Generated Content (AIGC). Current AIGC detection methods struggle to adapt to interactive conversational settings, and lack the interpretability needed for diverse, non-expert stakeholders in customer service applications. We propose EMMM, a multi-dimension, multi-level, multi-strategy framework for explainable chatbot detection. EMMM balances latency, accuracy, and interpretability for diverse users. Experimental results show that EMMM provides explanations accessible to non-expert users, with 73% of survey responses preferring its outputs, while achieving accuracy comparable to state-of-the-art (SOTA) models and maintaining low latency, generating explanations within 1 second. Our code and dataset are open-sourced¹.

CCS Concepts

• Information systems → Decision support systems.

Keywords

AIGC Detection, Explainability

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¹<https://github.com/AngieYYF/EMMM-explainable-chatbot-detection>

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1 Introduction

Large language models (LLMs) have transformed human-AI interaction with their ability to generate human-like content [21] and conversational text [20]. However, this also enables misuse through large-scale user impersonation in online customer service or emergency response systems, risking contamination of retrieval index and compromised prioritization of genuine human needs.

Although AI Generated Content (AIGC) detection has been an active research area, most existing detectors focus on isolated text passages [8, 15, 22]. Dialogue introduces complex multi-turn dynamics where detection targets a single participant within an interleaved exchange. This creates an unusual input structure that necessitates specialized handling. Furthermore, the lack of human-centered interpretability in current detectors undermines trust among non-expert users with limited AI knowledge. Mainstream explanation methods exhibit notable limitations: local attributions [26, 29] are hard to interpret in the absence of ground-truth important tokens; globally aggregated explanations [18] often lack relevance to the target sample. These challenges constrain both the reliability and explainability of LLM chatbot detection in real-world applications.

We propose EMMM, an explanation-driven framework for interpretable AIGC detection in conversational settings. EMMM integrates multi-dimensional inputs (language and behaviors), operates at multi-level interaction (turn and dialogue), and employs multi-strategy explanation (local narrative and semi-global visual).

The main contributions of EMMM are as follows: (a) **Dialogue-aware**: models conversational structure and speech acts to support reliable and explainable online/offline chatbot detection. (b) **Efficient**: leverages a selector-predictor pipeline and offline preprocessing for sub-second explanation generation. (c) **Interpretable**: delivers non-expert user friendly narrative and semi-global visual explanations preferred by 73% of human survey responses.

2 Related Work

AIGC Detection. AIGC detection predicts whether a text was generated by a human or LLMs. Many methods have been developed. Binoculars [8] is a SOTA zero-shot method using two language models to compute an AI-likelihood score based on entropy differences, requiring no training data. In contrast, supervised models excel in specialized tasks by learning meaningful patterns from labeled data [2, 31]. Existing work primarily focuses on text passages,

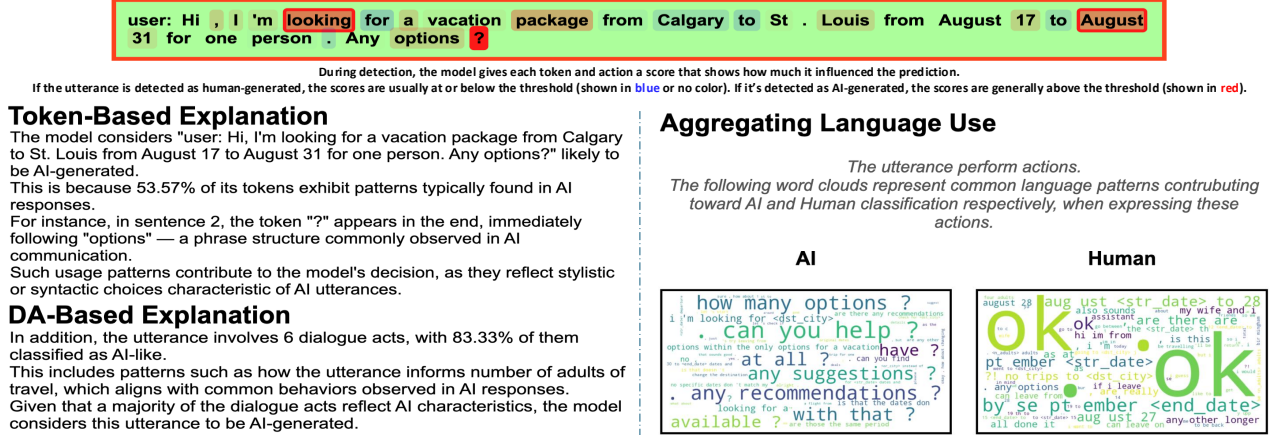


Figure 1: A demonstration of EMMM framework online detection and non-expert oriented explanation.

overlooking the complexity and interpretative richness of dialogues. To address this, we tailor the model to dialogue inputs and leverage dialogue-specific features to provide interpretable explanations.

Explaining AIGC Detection. Explaining a AIGC detection model involves understanding its decision-making process to foster user trust and evaluate model performance [14]. Two common approaches are attribution-based and Natural Language Explanations (NLE).

Attribution methods identify salient features for individual predictions [29]. This is the primary method used in existing AIGC detection work, assigning importance scores to extracted features [28] or input tokens [26]. This has been widely applied in tasks such as sentiment analysis, where attributions are expected to align with intuitively relevant input regions for interpretability [30]. However, AIGC detection lacks ground-truth important tokens, making interpretation more challenging. Global aggregation of local explanations can reveal broader model patterns to assist interpretation [18] but lacks relevance to individual predictions. We address this via a semi-global aggregation contextualized by target samples, enabling a higher-level yet relevant understanding of model behavior.

NLE methods produce textual explanations for human understanding. They can be generation- or template-based. The former produces explanations directly [14], but requires training with labeled NLE [16, 32]. NLE labeling is challenging for AIGC detection due to the limited understanding of the differences between human and LLMs in dialogues. Existing attempts targeted text passages and relied on domain experts, limiting scalability [10, 24]. Template-based approaches offer a more practical alternative, by defining explanation templates filled in per sample [33]. We propose a novel template using attribution insights to generate human-readable and interpretable explanations efficiently. This approach does not require prior intuition about the detection task for NLE annotation.

Explanations for Non-experts. Human-understandable explanations aim to convey a model's decision-making process to non-expert users, often through NLE [3] or intuitive visualizations [11]. Recent work has incorporated human-computer interaction (HCI) principles, with naturalness, flexibility, and usefulness emerging as critical factors [4, 10]. Our research focuses on these three aspects in designing explanation reports to better align with human values.

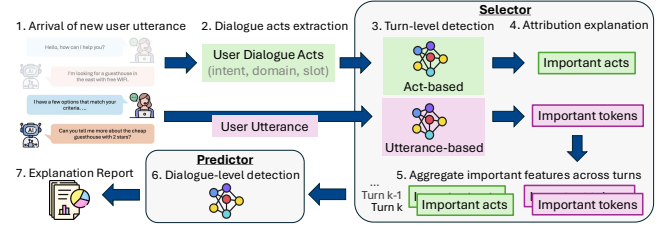


Figure 2: The 7 steps process of the EMMM framework.

3 Methodology

EMMM Framework. EMMM is a seven-stage framework for conversational AIGC detection (Figure 2). The three 'M's reflect its novel components of: Multi-Dimensional and Multi-Level detection and explanation, and Multi-Strategy explanation report.

Multi-Dimensional detection and explanation is grounded in the speech act theory that language not only conveys information but also performs actions through utterances [1]. Dialogue Acts (DAs) encode the speaker's intent and the pragmatic function of each utterance. For example, the DA (*inform, hotel, area*) indicates a user informing a hotel area preference. Existing approaches to AIGC detection and explanation predominantly rely on raw tokens, overlooking such structured communicative functions that offer a behavior-oriented perspective beyond surface-level linguistic cues. This motivates DA extraction to capture user intent and support richer Multi-Dimensional explanations besides linguistic signals.

Multi-Level detection and explanation addresses the incremental nature of multi-turn dialogues, where users expect real-time feedback. As feature attribution costs grow exponentially with accumulated turns, dialogue-level explanations become computationally expensive. To enable efficient explanation at both turn and dialogue-level, EMMM employs a sequential selector-predictor design [14]: the selector identifies important features from turn-level attributions, concatenated and passed as the dialogue-level explanation to the dialogue-level predictor. This eliminates the need to recompute the full explanation with each new utterance, and remains faithful by ensuring the predictor relies only on the provided explanations.

Algorithm 1 Semi-Global Aggregation - Offline Score Computation**Input:** Dataset $\mathcal{D}$ with utterances and Dialogue Acts (DAs)**Output:** Aggregated scores A , top features F

```

1:  $A, F \leftarrow \{\}, \{\}$  // Initialize scores and features
2: for each DA  $\in \text{DA\_types}(\mathcal{D})$  do
3:    $T_{\text{DA}} \leftarrow \text{ExtractFeaturesPerDA}(\mathcal{D}, \text{DA})$ 
4:    $L \leftarrow \text{GetLocalAttributionScores}(T_{\text{DA}})$ 
5:   for each class  $c \in \{\text{AI}, \text{Human}\}$  do
6:      $A[\text{DA}][c] \leftarrow \text{GlobalAggregation}(L, c)$ 
7:      $F[\text{DA}][c] \leftarrow \text{TopK}(A[\text{DA}][c])$ 
8: return  $A, F$ 

```

Multi-Strategy explanation report integrates two complementary strategies to enhance the understanding of non-expert users: local narrative explanations and contextualized semi-global visual insights. An example explanation report is shown in Figure 1.

These components are operationalised through the following seven-steps process (Figure 2): (1) a new utterance is received; (2) its DAs are extracted; (3) turn-level detection is performed using both the utterance and its associated DAs; (4) salient DA and token features that inform the turn-level prediction are identified via attribution; (5) these salient features are accumulated across turns to construct a compact dialogue-level representation; (6) dialogue-level detection is performed using this aggregated representation; and (7) a multi-strategy explanation report is generated.

Narrative Explanations. Inspired by HCI principles for explainable AI [5] and discourse analysis techniques [13], we design a lightweight natural language template with low computational complexity following three criteria: naturalness (how easily users can comprehend the model’s output), flexibility (ability to convey multiple perspectives and information types), and usefulness (effectiveness of the explanation to resolve users’ potential confusion about detection results). Discourse-aware elements, such as highlighting rhetorical patterns, help bridge the gap between technical attributions and non-expert understanding. The template was iteratively refined with expert inputs and user feedback to ensure clarity and accessibility without compromising efficiency.

Contextualized Semi-global Aggregation. We propose Contextualized Semi-Global Aggregation to address the limited relevance of global model insights towards specific target samples. A feature refers to a token or phrase extracted from text spans associated with a specific DA². During the offline phase (Algorithm 1), local attribution scores are computed for each token in the dataset, and phrase-level scores are obtained by averaging the attributions of constituent tokens. Features are grouped by DA type, and global aggregation methods are applied within each group to produce DA-specific attribution profiles for each class. The top ranked features per DA for the AI and Human classes are recorded with their associated scores. During online application to a target utterance (Algorithm 2), semi-global important features are identified by accumulating scores of the top features for each DA in the utterance. This DA-aware aggregation balances the interpretability of global model insights with the contextual relevance of local explanations.

²Based on embedding similarity using a fine-tuned paraphrase-MiniLM-L6-v2.**Algorithm 2** Semi-Global Aggregation - Online Explanation**Input:** Target DAs $\mathcal{D}\mathcal{A}_{\text{utt}}$, target class c , top features F **Output:** Semi-global important features S

```

1:  $S \leftarrow \{\}$  // Initialize feature-score map
2: for each DA  $\in \mathcal{D}\mathcal{A}_{\text{utt}}$  do
3:   for each  $(f, s) \in F[\text{DA}][c]$  do
4:      $S[f] \leftarrow S[f] + s$  // Accumulate score
5: return  $S$ 

```

Table 1: Comparison of offline detection Macro-F1. Binoculars¹ uses the default threshold of 0.9015, whereas Binoculars² uses tuned thresholds based on a validation set .

Category	Detection Model	SPADE	Frames	Average
Zero-shot	Binoculars ¹	0.5361	0.8115	0.6817
	Binoculars ²	0.8272	0.8191	0.8232
Pre-trained	ChatGPTD _{roberta}	0.5787	0.3989	0.4888
	RADAR	0.3452	0.4738	0.4094
Supervised	Entropy	0.5990	0.6539	0.5572
	Raidar _{llama}	0.7471	0.7945	0.7708
	Random Forest	0.9733	0.9902	0.9818
	MLP	0.9906	0.9976	0.9941
& Explainable	EMMM (ours)	0.9771	0.9945	0.9858

4 Experimental Evaluation

Evaluation Settings. SPADE [12] is the only existing benchmark, with hotel-domain bona fide and LLM-generated dialogues. We use its E2E Convo. dataset, and expand domain coverage by extending the data generation framework to the travel-domain Frames dataset [27] using Qwen2.5-32B [23], resulting in 1364 pairs of bona fide and synthetic dialogues. We randomly divide each dataset by dialogue ID into train, validate, and test sets in a 70%/15%/15% ratio.

The EMMM implementation fine-tunes a DistilGPT2 model [25] for turn-level detection models. Faith-SHAP [29] identifies up to three DAs and three tokens per utterance with the highest absolute attribution scores. The turn-level detection models are further fine-tuned for dialogue-level detection, where act and token embeddings are combined via average fusion and passed into the final classification layer. DA extraction uses an existing supervised model³ for SPADE, and few shot prompting Qwen2.5-7B [23] for Frames.

Detection Performance. Evaluated baselines include zero-shot Binoculars [8], pre-trained ChatGPTD_{roberta} [7] and RADAR [9], and supervised Raidar_{llama} [15], and Entropy detection [6, 12], random forest, and multilayer perceptron (MLP) with TF-IDF features.

During detection, user utterances are extracted from the dialogues and concatenated as detection input. Table 1 shows that zero-shot and pre-trained models, optimized for text passages, exhibit reduced efficacy on dialogues. EMMM achieves a competitive average Macro-F1 of 0.9858, outperforming zero-shot, pre-trained, and most fully supervised baselines. This shows that our proposed framework offers reliable detection performance besides advancing the explainability required for human-centered deployment.

Explanation Relevance. We use two aggregation methods: AGG [18] and LOR [17]. $AOPC_k(G, c)$ [18] quantifies the impact of the top k tokens from aggregation G on class c predictions. Higher

³<https://huggingface.co/ConvLab/t5-small-nlu-all-multiwoz21-context3>.

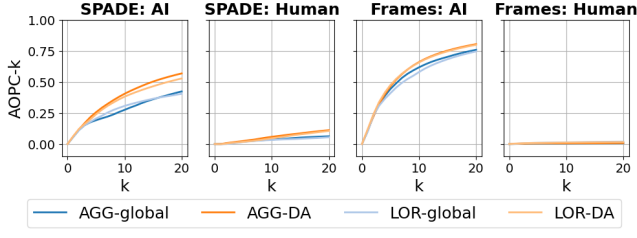


Figure 3: Comparison of explanation relevance between semi-global (-DA) and global (-global) aggregations using $AOPC_k$ scores (y-axis) across different values of k (x-axis). Semi-global aggregation (-DA) consistently outperforms global aggregation (Wilcoxon signed-rank test, $p < 0.05$).

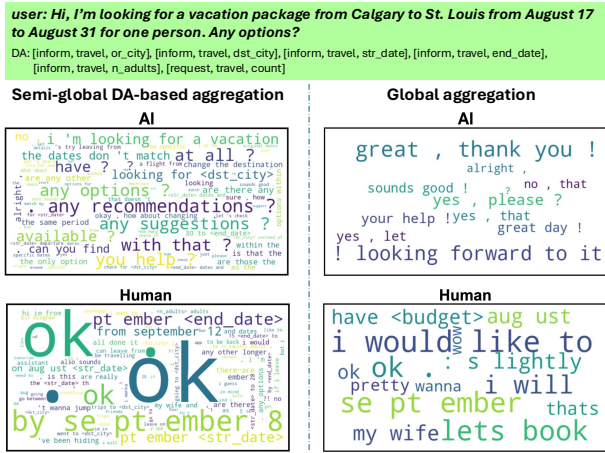


Figure 4: Illustration of semi-global and global aggregation. Semi-global highlights more utterance-relevant phrases.

scores reflect stronger relevance and alignment with model behavior. Figure 3 compares AOPC results between the proposed DA-based semi-global and global aggregations. DA-based aggregation achieves consistently higher scores, with one-sided Wilcoxon signed-rank tests on paired per-sample $AOPC_k$ values across $k \leq 20$ yielding $p < 0.05$ in all settings. The lower AOPC scores for human-predicted samples aligns with prior studies showing that human language exhibits greater linguistic variability [19], which reduces the overlap between top aggregated tokens and those in individual samples. Overall, these results indicate that our contextualization technique based on DA offers semi-global model insights that are more relevant and informative for understanding local predictions.

For qualitative analysis, Figure 4 illustrates DA-based and global aggregation, showing the top 20 phrases per DA and globally, ranked by frequency-weighted LOR. DA-based aggregation provides utterance-relevant explanations: AI features include question-oriented phrases like ‘any options?’ and ‘any recommendations?’ for travel needs, while human features convey needs directly and may include personal context such as ‘my wife’ or ‘friends’. In contrast, global aggregation highlights the most influential class-specific phrases across the dataset but lacks utterance specificity.

Interpretability. We conducted a human survey to evaluate the interpretability of EMMM’s output explanations. We randomly selected four representative samples, each covering a different dataset and label. 30 participants reviewed these samples, resulting in 120 survey responses indicating their preference between EMMM explanation and a baseline explanation. The baseline is the content of local feature attribution methods, the primary existing approach for explaining AIGC detectors [26, 28]. It displays the sorted top 10 important tokens and 3 DAs per utterance with their attributions.

Our explanation framework offers general advantages across user groups, particularly in enhancing accessibility and interpretability for non-technical users. 73% of the responses preferred EMMM explanations over the baseline. 12 participants without prior knowledge of AI models showed an even stronger preference for our framework (75%), compared to the 18 participants with prior knowledge (72.22%). This highlights its potential for real-world applications requiring explanations interpretable by a broad audience.

To assess the alignment of explanations with the HCI principles introduced in Section 3 (naturalness, flexibility, and usefulness), sixteen participants were further asked to provide 5-point scale ratings on these three criteria. The baseline achieved an average rating of 2.77, whereas EMMM achieved 3.81, demonstrating strong adherence to human-centric explanation design principles.

Time Efficiency. We measure average runtime over 100 randomly selected utterances per dataset. The 7-step framework runs in 2.19s on SPADE and 4.70s on Frames. Report generation is highly efficient (0.41s) using offline aggregation and natural language templates. The full explanation process (attribution and report generation) completes in under one second, suitable for real-time use. While LLM-based DA extraction is the primary latency source for Frames (3.48s), ablation results show that replacement with a supervised T5-base model achieves high efficiency (0.29s) with robust downstream detection (F1 0.9970) and AOPC explanation relevance.

5 Conclusion

In this paper, we present EMMM, the first explainable chatbot detection framework tailored for LLM-generated content in conversational settings. EMMM addresses critical challenges in chatbot detection, including dialogue-specific structure and the interpretability gap for non-expert users. Through a dialogue-aware architecture and an efficient selector-predictor pipeline, EMMM achieves detection accuracy comparable to SOTA models, while delivering turn and dialogue-level explanations within real-time constraints. Our interpretability evaluation demonstrates that EMMM’s natural language explanations and semi-global visualizations significantly improve user comprehension, with a 73% preference over the baseline approach. These contributions mark an important step toward explainable and deployable AIGC detection systems for high-stakes domains such as emergency and customer service platforms.

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